



Generative artificial intelligence for de novo protein design

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Abstract

Engineering new molecules with desirable functions and properties has the potential to extend our ability to engineer proteins beyond what nature has so far evolved. Advances in the so-called ‘*de novo*’ design problem have recently been brought forward by developments in artificial intelligence. Generative architectures, such as language models and diffusion processes, seem adept at generating novel, yet realistic proteins that display desirable properties and perform specified functions. State-of-the-art design protocols now achieve experimental success rates nearing 20%, thus widening the access to *de novo* designed proteins. Despite extensive progress, there are clear field-wide challenges, for example, in determining the best *in silico* metrics to prioritise designs for experimental testing, and in designing proteins that can undergo large conformational changes or be regulated by post-translational modifications. With an increase in the number of models being developed, this review provides a framework to understand how these tools fit into the overall process of *de novo* protein design. Throughout, we highlight the power of incorporating biochemical knowledge to improve performance and interpretability.

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Introduction

Proteins are complex nanomachines that perform manifold biological functions, mediating regulatory networks, catalysing reactions at ambient temperatures, and forming materials with exceptional tensile strengths. Inspired by the varied chemical functionalities displayed by natural proteins, designing artificial proteins for specific applications has been a long-pursued objective in synthetic biology. Experimental methods such as phage display [1] and directed evolution [2] have been developed to identify or enhance proteins with desirable functions; however, they are limited by size and cost, and are constrained to use natural proteins as starting points. Therefore, computational strategies have been developed as alternatives to efficiently generate protein sequences with desired functionalities. Over recent years, there has been a rapid acceleration in the development of computational protein design, the most sophisticated approaches of which now vie with experimental techniques. The emergence and proliferation of these methods forms the central theme of this review (see [Table 1](#)).

Introduction to *in silico* protein design Protein space

De novo protein design is often conceptualised through the framework of protein space, an abstract landscape where every coordinate represents a protein sequence, with an additional coordinate signifying the protein's *fitness* or ability to perform a function ([Figure 1](#)). Navigating this vast landscape is the primary challenge of protein design, and the multitude of peaks and valleys that can trap a search process within suboptimal designs (the so-called *ruggedness* of the landscape) represent a traditional obstacle to generate novel proteins.

Traditional *de novo* protein design utilises physics-based methodologies with iterative search strategies, aiming to navigate protein space through modelling the biophysical interactions governing the designed sequences [3]. Tools, like Rosetta, developed based on these approaches have been successfully employed over the past 20 years to engineer numerous proteins *in silico* [4,5] (see Woolfson for a review [6]). However, these models are computationally expensive, often make assumptions that limit their predictive properties, and are prone to stalling at local minima, hindering the discovery

Model name	Ref	Controllable output	Model type (protein representation)	Release date (most recent update)	Maximum length	Experimental validation	Novel topologies	Is the code publicly available?	Is there a Google Colab tutorial or Hugging Face Space?
RefineGNN	[57]	Neutralisation of antibody CDRs	Autoregressive GNN	27/01/2022	N/A	No	No	https://github.com/wengong-jin/RefineGNN	No
Anand Achim	[54]	Inpainting, secondary structure	Diffusion (frame representation)	01/05/2022		No	Yes	No	No
ProtGPT2	[26]	No	Autoregressive LLM	01/07/2022	250	No	Yes	https://huggingface.co/nferruz/ProtGPT2	https://huggingface.co/nferruz/ProtGPT2
ProteinSGM		Scaffolding	Diffusion (descriptive square matrix representation)	01/07/2022	256	Yes	Yes	https://gitlab.com/mjslee0921/proteinsgm	No
ProGEN2	[27]	Enzyme family tags	Autoregressive LLM	01/08/2022	512	Yes (~100 proteins)	Yes	https://github.com/salesforce/progen	No
FoldingDiff		Length of desired protein	Diffusion (angle representation)	01/09/2022	128	No	N/A	https://github.com/microsoft/foldingdiff	https://huggingface.co/spaces/wukevin/foldingdiff
ESMDesign	[14]	Backbone shape	Masked LLM	01/12/2022	184	Yes (228 proteins)	Yes	https://github.com/facebookresearch/esm/tree/main/examples/lm-design	https://colab.research.google.com/github/facebookresearch/esm/blob/main/examples/protein-programming-language/tutorial.ipynb
ZymCTRL	[28]	Enzyme family tags	Autoregressive LLM	01/12/2022	600	Announced but not yet provided	Yes	https://huggingface.co/nferruz/ZymCTRL	https://huggingface.co/nferruz/ZymCTRL
RFDiffusion	[13]	Length, shape, scaffolding, binding partners, oligomerisation	Diffusion (frame representation)	01/12/2022	600	Yes (~1000 proteins)	Yes	https://github.com/RosettaCommons/RFdiffusion	https://colab.research.google.com/github/sokrypton/ColabDesign/blob/v1.1.1/rf/examples/diffusion.ipynb
Chroma	[15]	Length, shape, scaffolding, binding partners, natural language annotations, oligomerisation	Diffusion (frame representation)	01/12/2022	600	Yes	Yes	https://github.com/generatebio/chroma	No
DiffSDS	[38]	Inpainting	LM + Diffusion (direction representation)	01/01/2023	30 inpainted residues	No	N/A	https://github.com/A4Bio/DiffSDS_open	No
Genie		No	Diffusion (frame representation)	01/02/2023	128	No	Yes	https://github.com/aqlaboratory/genie	No
ProteinDT	[16]	Function using natural language annotations	LM + Diffusion (sequence representation)	01/02/2023	N/A	No	N/A	No	No
ProtSEED	[58]	Secondary Structure		02/03/2023	N/A	No	No	https://github.com/shichence/ProtSeed	No
FrameDiff		Length		01/05/2023	500	No	Yes		No

Protein Generator	[37]	Amino acid composition, length, scaffolding, introduction of cleavage sites	Diffusion (frame representation)				Yes	Yes	https://github.com/jasonkyuyim/se3_diffusion	https://huggingface.co/spaces/merle/PROTEIN_GENERATOR
Protpardelle	[56]	Motif scaffolding (all atom) and inpainting	Diffusion (point cloud)	25/05/2023	300	No	Yes	Yes	https://github.com/alexechu/protpardelle	https://huggingface.co/spaces/ProteinDesignLab/protpardelle
AbODE	[59]	Antibody CDR generation	Graph Neural ODE	31/05/2023	15–20	No	No	No	No	No
EvoDiff	[36]	MSA families, IDRs, motif scaffolding	Diffusion (sequence representation)	12/09/2023	N/A	No	Yes	Yes	https://github.com/microsoft/evodiff	No
RFDiffusionAA	[21]	Protein binders to small molecules	Diffusion (frame representation of proteins)	09/10/2023	150	Yes	Yes	Yes	Not as of publication	Not as of publication

of diverse functional variants. Ultimately, understanding the intricacies of protein physics remain one of the central problems in modern biochemistry, and as physics-based designs methods near the boundaries of our understanding, their capabilities become increasingly limited.

Generative models

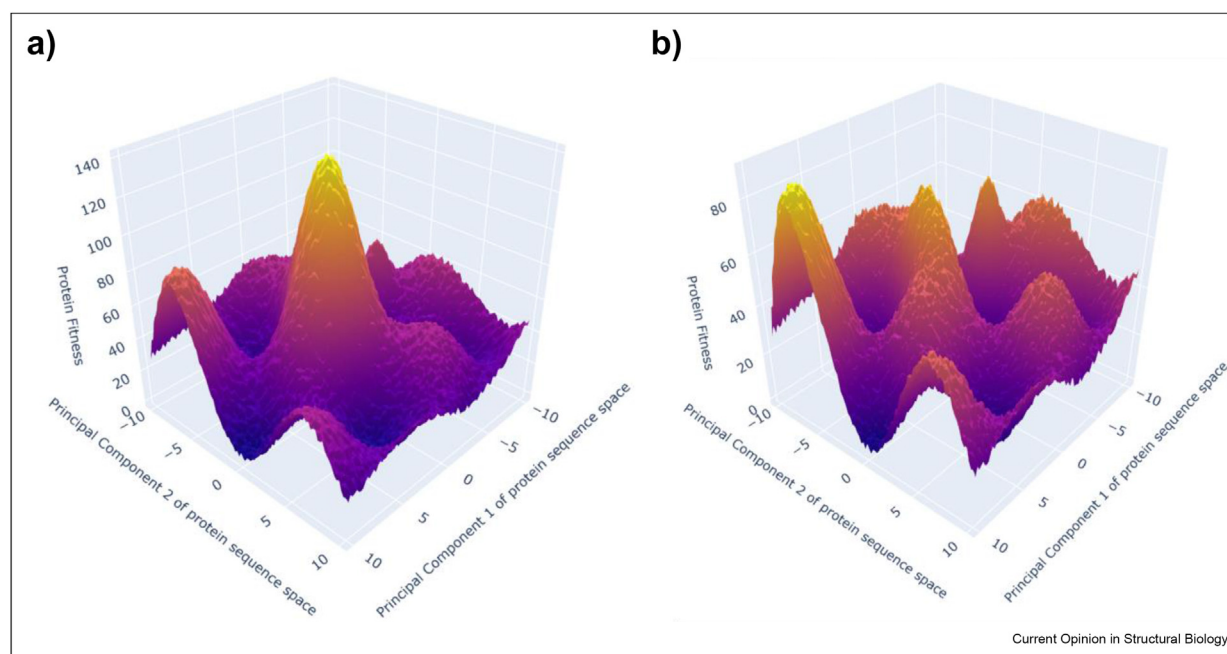
An alternative approach to physics-guided protein design involves selecting candidates with desired properties exclusively from ‘plausible’ regions of protein space, signifying proteins that have a reasonable likelihood of performing specific functions of interest. This search can be conducted by *training* a generative model — a statistical model that learns a distribution over protein space. As the model can simultaneously model the variables that determine protein sequence and fitness, it is then possible to ‘sample’ protein instances from regions of high fitness for a particular task.

The advent of deep learning has enabled learning generative models. In this review, we organise the latest *ab initio* design techniques into two groups: sequence-based and structure-based approaches. **Structure-based design has recently focused on the use of diffusion models (the models behind text-to-image generators [7–10]), although with a significant contribution from graph neural networks (GNNs), whereas sequence-based design has primarily focused on large language models (LLMs) ([11]).** As we will argue, these models have learned to generalise beyond the parts of protein space inhabited by natural proteins and can venture areas of this space unexplored by nature [12,13]. Moreover, they are enabling scientists to specify desired properties using easy-to-use programming languages [14] or even natural language [15,16]. Recent advances in improving *in silico* checks might further increase the success rate of designed proteins to close to 100% [17].

Protein structure prediction

Protein structure prediction algorithms (which we term sequence-to-structure algorithms, Figure 3.), such as AlphaFold [18], RoseTTAFold [19,20], RoseTTAFoldAA [21], and ESMFold [22] have particularly benefited *de novo* protein design for two reasons. First, they have provided *in silico* metrics to score predicted designs. Bennett et al. [18] discovered that filtering designs based on the metrics ***predicted Local Distance Difference Test* (pLDDT — a per-residue model confidence measure) and *Root Mean Squared Deviation* (RMSD) between the coordinates of atoms in the structures from the generative model and coordinates predicted by a structure prediction algorithm on the sequence of the designed protein, increased experimental success rates tenfold.** Second, they have provided an abundance of structural training data for other models. For example, Hsu et al. [23] used 12

Figure 1



Example of protein fitness landscapes. a) This protein fitness landscape has clear local maxima with high fitness **b)** A more rugged protein fitness landscape with many maxima and minima that make it hard to navigate.

million protein structures predicted by AlphaFold2 to train the structure-to-sequence model ESM-IF.

One current limitation with their use in protein design is their adaptability to novel topologies absent from training sets. These methods also exhibit limitations in predicting physiologically relevant protein states or components. For instance, they cannot yet model protein conformational landscapes [24], and whilst RFAA can now model ligands, co-factors and post-translational modifications (PTMs), knowledge of the combination of PTMs present under different conditions still requires experimental determination. Structure prediction methods are based, by and large, on deep neural networks trained on crystal structures, and as such they may be considered akin to structural studies of proteins, which while insightful are also limited to specific states of the proteins that may not well represent how they behave in the cell.

Sequence-based protein design

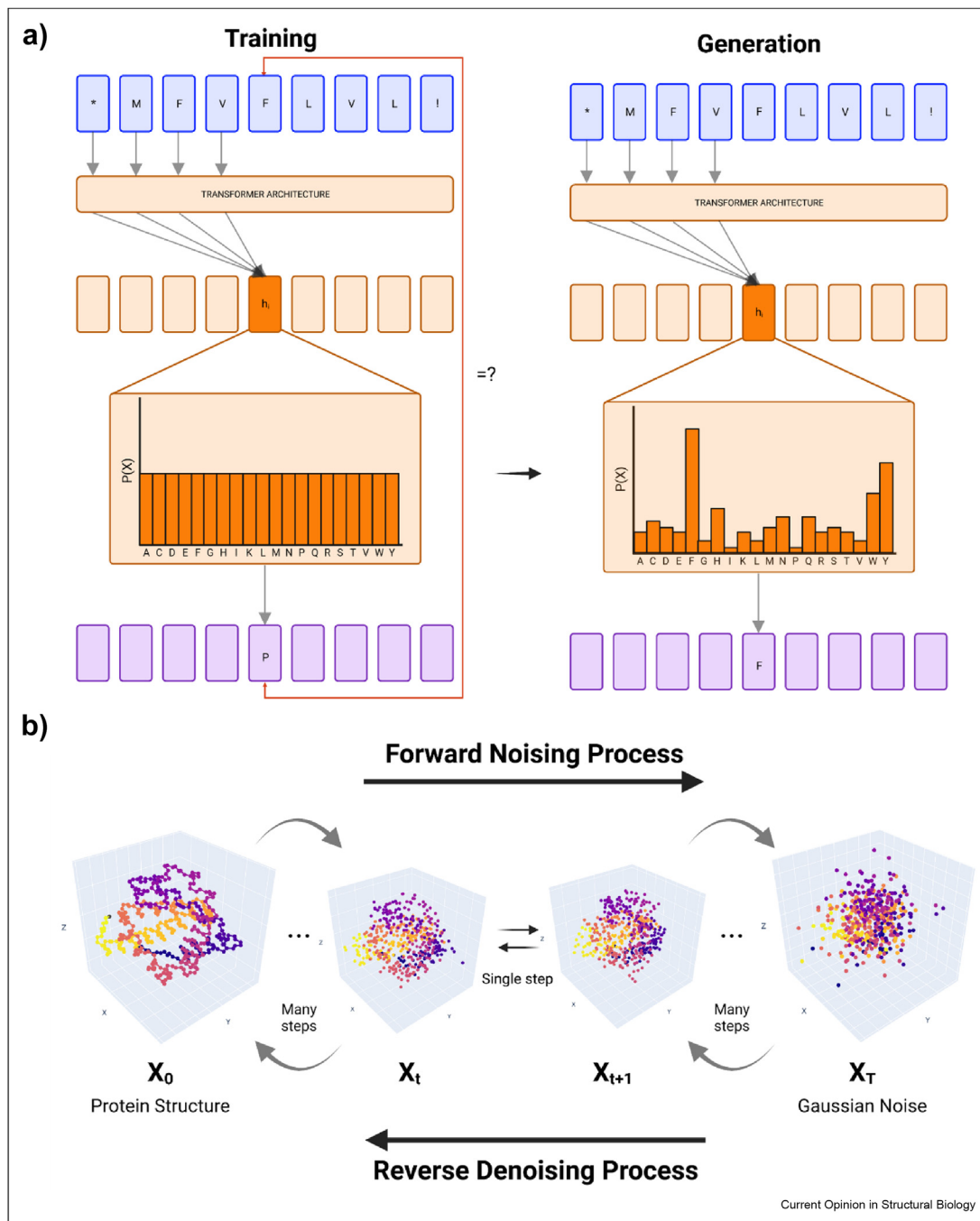
We first discuss sequence generation (properties-to-sequence algorithms, Figure 3.) which primarily uses *large language models* (LLMs). The way LLMs have addressed the challenge of sampling from plausible protein space can be thought of as learning the grammar of protein language (see Figure 2a and Box 1). Models are trained to predict the identity of masked amino acids

when given the context of some parts of a protein sequence [25]: during training, the model's predictions are compared to the known sequence and its weights are then updated based on whether it made a correct or incorrect prediction. When asked to generate new proteins, models typically predict amino acids sequentially to build up a novel protein sequence. Increasingly, these sequences have both low homologies to known proteins, similar physicochemical properties to natural proteins [26], and high experimental success rates.

One of the first examples of protein design using language models was Verkuil et al.'s model [12] which generated proteins incorporating complex structural motifs, such as hydrogen bond networks. Of the 152 successful experimental designs (out of 228 tested) in Verkuil's models, 35 had no significant sequence match to known natural proteins, and the remaining 117 had a median sequence identity of 27% to their nearest matches. **ProGEN** [27] also demonstrated an experimental success rate comparable to controls set of random and natural proteins. Furthermore, Verkuil's model, **ProtGPT2** [26], and **ZymCTRL** [28] also demonstrated low *structural* identity to known proteins.

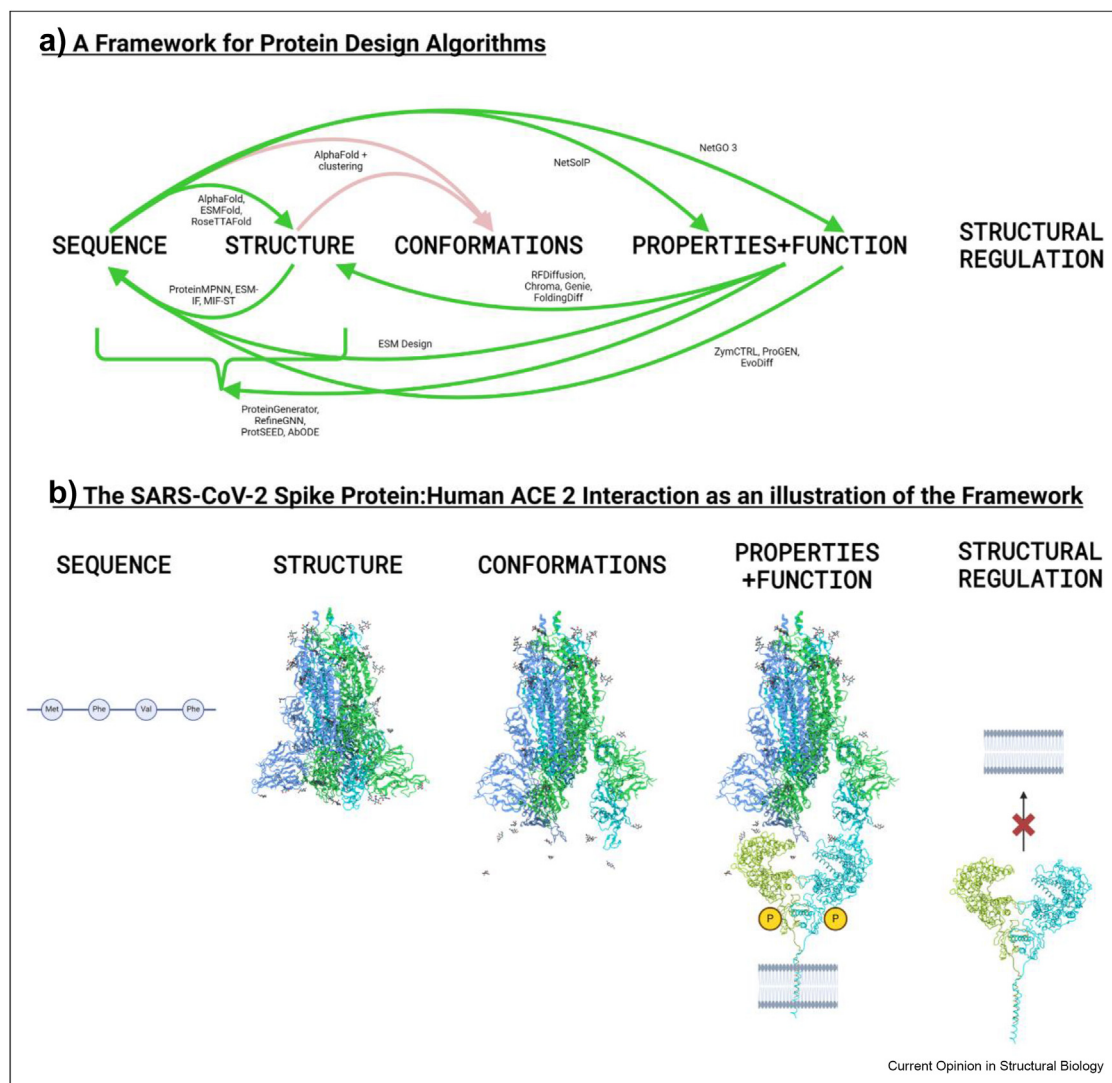
Large language models can also generate protein sequences with a specific function by utilising conditioning tags or Markov Chain Monte Carlo (MCMC)

Figure 2



The training and generation schemes for language and diffusion models **a)** Language models are trained on large sets of protein sequences without any explicit guidance on what patterns to look for. The training process involves presenting the model with a protein sequence and asking it to predict a masked amino acid in the sequence, with the exact context provided depending on the model architecture (see Box 1). When the model makes a prediction, its weights are updated based on whether it made a correct or incorrect prediction (shown by the red arrow). Once trained, the model generates new protein sequences based on a given context by sampling from its learned probability distribution, which assigns probabilities to each possible next amino acid. **b)** Diffusion models are trained by taking known protein structures and noising the chosen representation of the protein (typically atomic coordinates) by applying a diffusion process which adds Gaussian noise. A neural network is then trained to learn the reverse of the noising process of known proteins (see Box 2). Once trained, the model can be provided with true Gaussian noise and generate from this realistic protein structures. Figures adapted from (a) [44] (b) [22] and created with Biorender.

Figure 3



a) A framework to understand the inputs and outputs of computational models developed for protein design. Proteins are polymers of amino acids, forming a sequence that folds into a 3D structure. Here structure represents a single low-energy conformation, while conformations account for the entire ensemble of accessible shapes. Structural regulation represents switching on or off protein activity by addition or removal of chemical groups to the protein. Green arrows indicate rapidly advancing algorithm development. For simplicity, where there are gaps in the framework to be filled in, red arrows are used to signify tractable next steps for algorithm advancement. **b)** SARS-CoV-2 spike protein's amino acid sequence folds into a 3D structure, forming a trimer of three identical chains. The spike protein adopts different conformations, transitioning between closed and open states. The open state enables binding to human ACE2 receptors. ACE2 receptors exhibit structural regulation, with phosphate group removal reducing their cell surface presence. PDB IDs: 7DWY, 7WS8, 6M17. Figure created with Biorender.

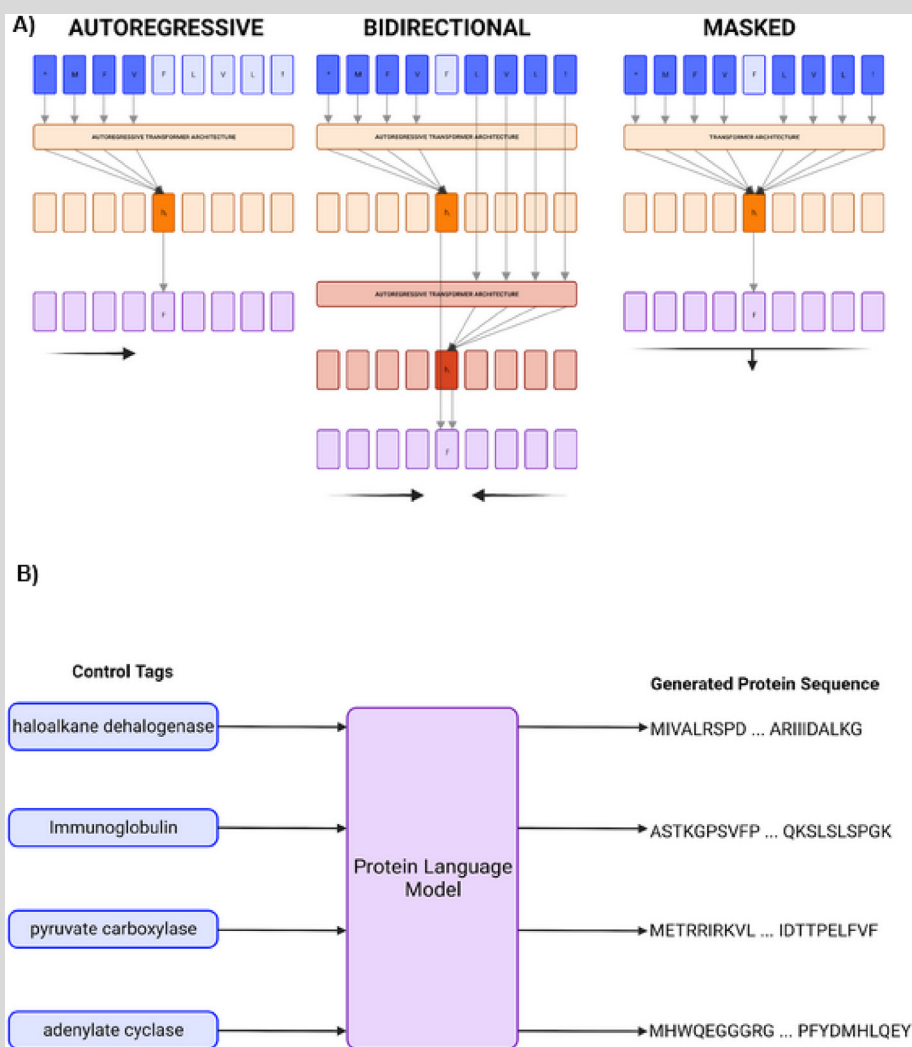
searches (see [Box 1](#)). For example, during the training of ProGEN and ZymCTRL, enzyme database classification codes were attached as tokens to the start of the protein sequence. This enabled the models to learn different probability distributions for different enzyme families. When ProGEN was used to generate proteins with 'lysozyme' activity, 73% of the designs were functional.

Language models have also played a leading role in protein engineering, where the objective is to optimise some properties of a known protein. **LLMs are excellent**

starting points to build models that can predict protein properties (which we denote as seq-to-property models, [Figure 3](#)). The intermediate states in the language models (known as *latent variables* or, more commonly, *learned representations*) can be used as inputs for a variety of machine learning models. These representations are central to the state-of-the-art tools for predicting numerous protein properties [29]. In turn, these models can be used in sophisticated pipelines to optimise the properties of a starting protein (see reviews by Hie et al. [30] and Yang et al. [31]).

Box 1. Language model architectures and functional conditioning approaches. Figures adapted from (a) [44] and (b) [45].

(A) Large language models (LLMs) utilise self-supervised learning to generate learned representations of proteins that capture relationships between various amino acids. The training process involves presenting the model with a protein sequence and asking it to predict a masked amino acid in the sequence based on only the previous amino acids (autoregressive language models, e.g. GPT), all other amino acids (masked language models, e.g. BERT), or the previous and following amino acids independently (bi-directional language models) [28]. When the model makes a prediction, its weights are updated based on whether it made a correct or incorrect prediction. This process is repeated many times with the models being trained to minimise the negative log-likelihood over the dataset $D = \{x^1, \dots, x^{|D|}\}$. **(B)** Strategies to provide control over generated sequences' properties include: 1) high-throughput generation and filtering of sequences with desired predicted properties; 2) incorporating conditioning tags (shown); 3) employing prompt-engineering, as with RITA; and 4) fine-tuning on specific families, as in ProGEN; 5) Adopting an MCMC search approach, whereby a language model iteratively proposes mutations to an initial sequence and accepts or rejects them based on an energy function encoding desired properties and constraints. Over many iterations, sequences evolve toward low energy, meeting specified design criteria.



Despite these early successes, limitations remain. New structures with novel catalytic mechanisms or entirely new functions previously designed rationally [5,32,33] are yet to be created using language models. Furthermore, search methods like MCMC can be time-consuming due to the extensive number of iterations

required and their tendency to become trapped in local minima.

To overcome some of these limitations, an alternative sequence-based design methodology has been developed to utilise diffusion models, discussed further in the

next section, on a text-based task [34,35]. The first sequence diffusion model to be trained on the entire UniProt sequence space was **EvoDiff** [36], which leveraged its diffusion objective to generate intrinsically disordered regions of proteins — a task intractable for structure-based techniques due to the impossibility of obtaining single structures. Protein Generator [37] and DiffSDS [38] are also sequence-based diffusion models; however, these are discussed in the next section as they are also trained using structural information.

Structure-based protein design

Structure-based design methods focus on creating novel protein structures (properties-to-structure algorithms, Figure 3.). We will focus our discussion on diffusion models, a deep generative model that has shown promise for protein structure generation, although with mentions to two non-generative structure-based techniques, hallucination and reinforcement learning. We first introduce inverse folding (structure-to-sequence models, Figure 3.) as an essential technique for structure-based design that enables sequence generation from structural models.

The advent of rapid and accurate protein structure prediction enabled design methods like *hallucination* [39,40] (reviewed in Ref. [41]), where the structure predictor is subject to a search algorithm, typically MCMC, to find sequences that fold onto a desired shape. While this technique is subject to ongoing improvements [42], it does not generate a probability distribution over protein space, it is computationally slow, and it is limited to designing a scaffold around well-defined sites. A similar strategy **used reinforcement learning in combination with Monte Carlo Tree Search to iteratively design proteins in a ‘top-down’ approach** [43]. Whilst successful in designing nanopores and ultracompact icosahedra, it also is not truly generative, in the sense that it does not model a joint probability distribution over protein coordinates and fitness.

Inverse folding

Generating sequences that fold into a given structure — termed *inverse folding* by the community and which we refer to as structure-to-sequence algorithms — constitutes a crucial aspect of numerous structure-based de novo protein design methodologies [3]. Initial approaches relied on physics-based tools, like Rosetta, to generate sequences. Recently, graph-based deep learning techniques (also known as *geometric deep learning*) have proven remarkably adept at this task [23,44–49]. Graph neural networks conceptualise each backbone atom of protein's structure as a node on a graph, and connections are drawn to all other atoms within a defined distance, forming the graph's edges. Each node and edge can be provided with additional information: nodes can be annotated with their amino

acid value or 3D coordinates; and edges can be annotated with vectors that describe the distance, direction, and orientation between the nodes they connect. These three specific edge features, first introduced by Ingraham et al. [44], importantly are invariant to rotations and translations of the protein and as such have been used and built on by many more recent models (Box 3). Within the model's architecture, information can be passed between neighbouring atoms to learn the local environment in two ways: the first, known as message passing, performs operations on all of a node's edges context features, and the second is to use attention mechanisms, as in language models. Relating these ideas to structure-to-sequence protein design, was again first introduced by Ingraham et al. [44] to create structurally-informed representations of the protein. **The success of graph neural networks for structure-to-sequence algorithms is exemplified by their use to redesign previously unsuccessful protein structures, increasing experimental success rates [45].** Tools like ProteinMPNN have repeatedly improved the yield of soluble protein from initially unsuccessful designs [50–53]. Several other graph-based structure-to-sequence techniques exist, including ESM-IF [23], PiFold [47], MIF-ST [46], and Knowledge-Design [49].

The ability to produce sequences that fold to a given protein structure is not limited to single structures. For example, the generative process can be biased to generate protein assemblies with desired stoichiometries and symmetries [13,39].

Diffusion models

Structure-to-sequence methods require an initial protein structure that can be successfully translated into a sequence. In the past year, significant interest has focused on *diffusion models*, a generative deep learning architecture that sidesteps this limitation. Diffusion models [7–9] can be thought of as learning the process of drawing proteins (see Figure 2b and Box 2). **During model training, experimentally determined protein structures are noised by iteratively adding small amounts of Gaussian noise. The model is then trained to recover the original protein structure from the noised data through an iterative denoising process (see Box 2 for details). A loss function on the difference between the true denoised protein structure or true gradient of the reverse diffusion process at a given time step is then used to train and update the weights of the neural network.** This enables the model to learn probability distributions of protein space such that, at generation time, the model can be provided with true Gaussian noise and iteratively generate novel protein structures.

Initial success was achieved by diffusion models that generated backbone coordinates to produce diverse monomers and protein assemblies [54]. Proteins

Box 2. Diffusion model architectures and functional conditioning approaches.

Diffusion models, a class of thermodynamics-inspired generative models, have their roots in the initial concept introduced by Sohl-Dickstein in 2015. An analogous concept, known as Score Matching with Langevin Dynamics, was independently developed by Song in 2019. Ho *et al.*, 2020, built on the concept introduced by Sohl-Dickstein to implement them for generating high-quality images in a form they called Denoising Diffusion Probabilistic Models. Subsequently, these models were brought together under the umbrella of score-based generative modelling by Song in 2021. Advancements in 2022 saw Karras *et al.* providing a modular framework for diffusion models and Ho *et al.* introducing progressive distillation as a novel training technique. In 2023, Song further contributed to the field by introducing consistency models. Concurrently, a unique electrostatics-inspired generative model, termed Poisson Flow Generative Models, was proposed by Xu *et al.*, in 2022. In 2023, these authors extended Poisson Flow Generative Models into higher dimensions and integrated an augmentation dimension parameter, which surprisingly enabled the unification of these models and diffusion models under a single model family.

A. The noise induction process of a diffusion model applied to a basic 1D probability distribution function. Here, samples are drawn from a probability distribution, then Gaussian noise is progressively added by modelling a thermodynamic diffusion process. The resulting distribution closely approximates Gaussian noise.

B. Each step in the time process generates a new noisy data distribution. Diffusion models can be viewed as performing multiple data augmentation steps as part of the noising process, which is in contrast to traditional deep learning methods that typically resort to a few transformations such as translations and rotations. As the variable 't' approaches 'T', the distribution increasingly aligns with a Gaussian.

C. A neural network model is trained to 'denoise' the data by predicting the reverse of the noising process, which is deterministic, across many different samples. Training starts with samples drawn from the unknown x_0 distribution. A forward noising trajectory of that sample is created and then a time point 't' is randomly sampled from a uniform distribution of [0,T]. Using Bayes' rule, it is possible to obtain x_t directly from x_0 . Then a neural network, known as the score function, is trained to either directly retrieve x_0 from x_t (as per Karras *et al.*) or to predict the gradient of the denoising trajectory at time 't' (as per Song *et al.*). A loss function is then either applied to the difference between the predicted x_0 and the true x_0 or between the predicted gradient of x_t and the true gradient of x_t . The true gradient can be obtained from the reverse of the noising process, which is deterministic and can be calculated from the forward noising process when x_0 is known. This training is iteratively performed over multiple timesteps, noise induction trajectories, and samples. In Denoising Diffusion Probabilistic Models (DDPMs) as introduced by Ho *et al.* in 2020, a different loss function of the difference between the true and predicted noise at different time steps given x_0 is used, which derives from a comparison of the true and predicted x_t distributions. Once trained and having learnt to predict the reverse of the noising process, a model is then capable of predicting such a denoising trajectory from random noise and hence generate samples (images or proteins) that fit to the training x_0 distribution. Predicting a denoising trajectory from random noise can be referred to as a *generation* or *generative sampling process*.

D. Diffusion models can be guided towards specific design objectives using Bayesian Inference during each step of predicting a denoising trajectory from random noise. To condition on a property, y , a time-dependent classifier model $p_t(y|x)$ is trained on the noised structures $x_t \sim p_t(x|x_0)$. The generative sampling process can then be simply adjusted by adding the gradient of the log likelihood that structure x_0 will have the desired property at time $t = 0$ ($\nabla_x \log p_t(y|x)$) to the learned gradient from the diffusion model ($\nabla_x \log p_t(x)$).

In addition to classifier-based guidance, methods known as 'classifier-free guidance' have been developed. This technique avoids the complexity of training a separate classifier for conditioning. Instead, the diffusion model is jointly trained on a mix of conditioned and unconditioned data. During the generation phase, a conditioning strength parameter is adjusted to modulate the influence of the desired property on the generated samples (Ho and Salimans, 2022). This approach not only simplifies the model architecture but also provides flexibility in controlling the extent to which the property is expressed in the generated samples.

Diffusion models broadly segregate into two categories, those that incorporate stochastic elements in the denoising process (Stochastic Differential Equations, SDEs) and those that do not, relying on a specific reformulation of the reverse of the noising into an ordinary differential equation (known as the probability flow ordinary differential equation, PF ODEs). However, all these ODEs have the potential to introduce stochastic elements by adding noise (as per Karras, Chu). **The lighter shade of blue used here for the distribution of X_T represents that true Gaussian noise is now being used.**

E. During generation, the model is provided with true Gaussian noise. As per the training schedule, the model can either predict the gradient to guide the movement of the noised data or make a prediction of x_0 from the noised data and then compute the gradient based on the number of noising steps. The model then moves from the noised data to less noised data by moving the data in the direction of the predicted gradient. Further noise (ϵ) can then be injected at this point. In the diagram this is shown as the movement from point b to c. An alternative is to add noise before predicting x_0 and then compensating for the added noise by taking a larger denoising step. (a) Represents the predicted of x_0 . (b) Represents the movement from x_t to a less noised data distribution. (c) Represents the updated x_{t-1} with the injection of noise. The stochasticity can result in different generated structures from the same starting noise.

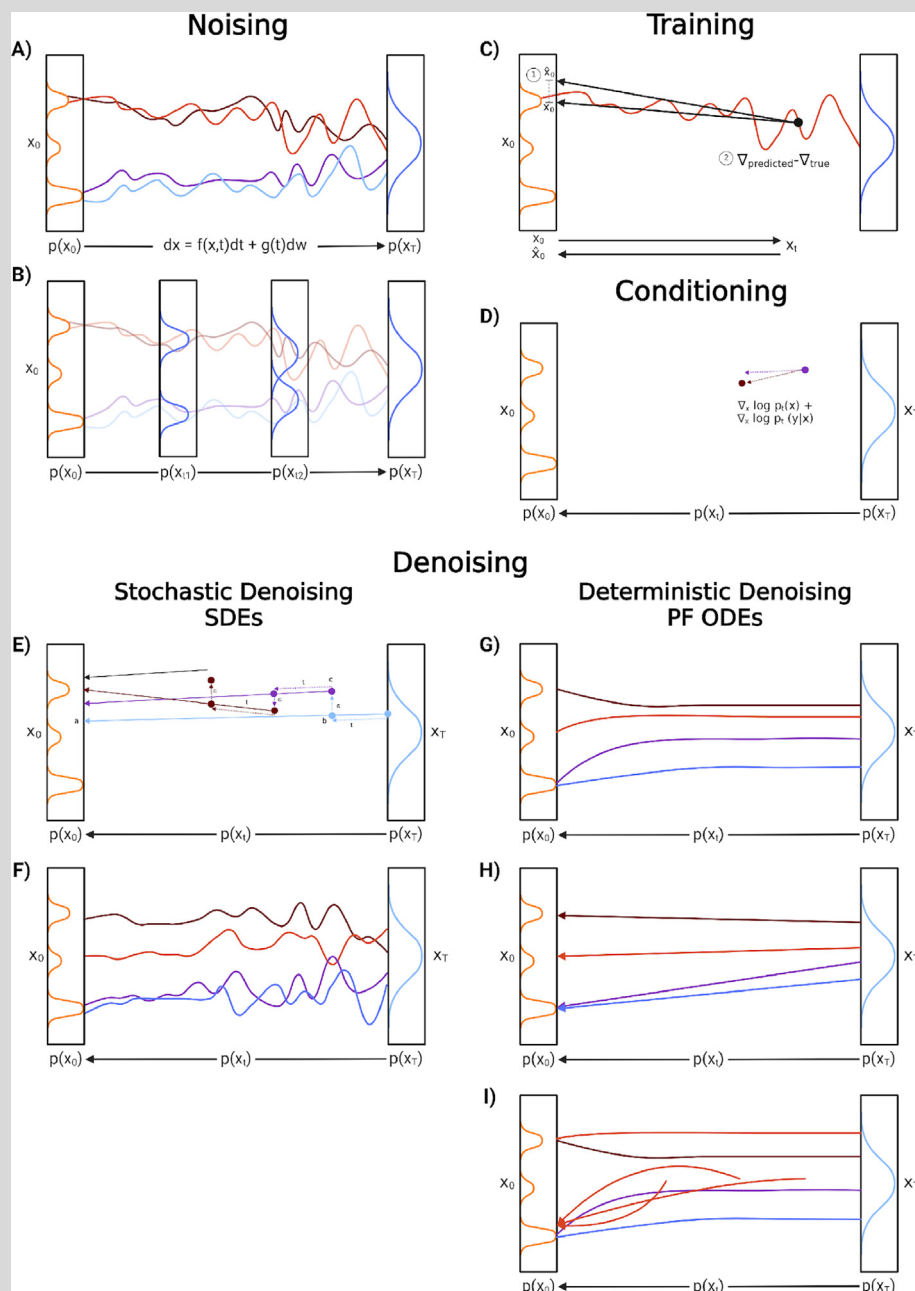
F. The solution trajectories for many denoising tasks. Surprisingly, diffusion models generate samples from the true x_0 distribution that were not present in the training samples (middle trajectory).

G. It is possible to obtain a deterministic ODE that matches the marginal evolution of the SDE. This approach results in more efficient and precise likelihood calculations, thereby improving generation.

H. Recent reformulations of the denoising trajectory (as per Karras et al.) set the noise step equal to time and do not scale the data, leading to more linear solution trajectories.

I. Consistency models propose a method to directly generate denoised samples from any point on the noising trajectory.

Box 2 diffusion model architectures and functional conditioning approaches.



generated by **Chroma** [15] contained a similar distribution of fragments of protein structures to natural proteins. **RFDiffusion**, which leverages RoseTTAFold's pretrained weights and ability to directly predict protein

structures, generated CryoEM-validated alpha-beta barrels architectures that had 16:16 and 12:24 helix:-strand ratios — extending beyond the structural variations of the classic 8-strand/8-helix TIM barrel fold

explored by evolution [13]. These models are backbone only and require a final structure-to-sequence step to map the structural model to a protein sequence.

Diffusion models can also be biased to incorporate desired properties and functions throughout the generation process (see Box 2, Fig D), like LLMs. The conditioning process results in quicker, more effective outcomes, and enables more semantic conditioning. For example, **RFDiffusion** was employed to generate *de novo* binders for five different targets, achieving an 18% success rate with 95 designs, and created binders to helical peptides with picomolar specificity [55]. **Chroma** and **ProteinDT** [16] can be further conditioned on natural language annotations. This introduces the tantalising potential for protein design queries to be articulated in natural language — for example, “Design a protein that binds to PD-L1, exhibits solubility, and exists as a monomer.”

Improvements to diffusion models

Diffusion models’ functional generation has largely focused on binder design and scaffolding without venturing extensively into designing proteins with novel enzymatic activity. A crucial element of enzyme design centres on finding the right arrangement of side chains to stabilise a substrate transition state. First-in-class diffusion models required sequential generation of backbones, sequences, and rotamers by structure-to-sequence and rotamer-packing modules meaning ligand-side chain and backbone-side chain interactions were largely ignored. However, newer generation diffusion models, especially those that incorporate side chains, offer significant promise in fine-tuning side chain positions for transition state stability.

One such all-atom diffusion model was developed by Chu et al. [56], which simultaneously designs sequence and structure. In their model, named **Protpardelle**, an all-atom protein structure is predicted at each time step and then the backbone coordinates of this model are used to predict a sequence using a modified structure-to-sequence module. The predicted sequence is subsequently fed into the next time step and all the atoms of the side chains of that sequence are diffused, with a diffusion size proportional to the last time the atoms appeared in the diffusion process. This allows the calculation of an updated final structure prediction encompassing all atoms but allows the sequence to change with each time step by utilising the backbone to propose a new sequence.

A related approach to Protpardelle is **RFDiffusionAA** [21] which also models all the atoms of ligands and post-translational modifications. This was achieved by first developing a modified RoseTTAFold that was trained on information about the chemical structure, bonding, and chirality of small molecules and PTMs bound to proteins.

As with RFDiffusion, a diffusion model was then trained using the network weights of this model. In contrast to Protpardelle the model does not generate every atom of the protein at every step in the diffusion process, rather a limited selection of residues from the protein are atomised into their atomic constituents. RFDiffusionAA was used to generate binders against digoxigenin (0.07% success rate), the heme cofactor (57% success rate), and the bilin pigments (3% success rate) involved in the light accumulation for photosynthesis.

Sequence and structure co-design

Models are increasingly breaking down the divide between exclusively sequence and structure-based design methodologies. For example, **RefineGNN** [57], builds upon Ingraham et al. [44] by using a structure-to-sequence model to autoregressively jointly design the sequence and structure of the binding motifs of antibodies. Starting from an initial antibody structure, binding motif residues are predicted one by one whilst iteratively predicting an updated global antibody structure at each step. **ProtSeed** [58] and **AbODE** [59] both improve on this by co-designing sequence and structure without iterative sampling. **ProtSEED** takes as input vectors that describe desired properties, such as secondary structure annotations and residue contacts, encodes these using a trigonometry-aware attention mechanism, and then uses a three-track neural network with cross-conditioning (representing sequence, backbone positions, and orientations), similar in structure to AF2, to predict sequence and structure together in a one-shot manner. **AbODE** uses a GNN to take as input a specific antigen structure and then uses a system of differential equations, which learn to directly simulate evolution of node and edge features over time rather than denoising as in diffusion models, to jointly generate antibody binding motif sequences and structures to bind the antigen. **Protein Generator** [37] is a diffusion model on sequence space that incorporates both structural and sequence information (Box 3).

Additionally, there have been attempts to incorporate more structural information into language models. For example, **SaProt** [60] leveraged the structural alphabet of **FoldSeek** [61] (an alphabet that provides each residue with one of 20 letters to describe its geometric conformation with its spatially closest residue) to create a protein language model that represented each amino acid as its type and its FoldSeek structural token. Although not used as a generative model in this case, it outperformed all previous language models on being able to predict properties of proteins and highlights the potential of incorporating more structural information into language models.

Designing dynamic proteins

A critical feature of proteins, enabling their diverse functional range, is their ability to change shape rapidly

Box 3. The importance of the representation of proteins**Sequence representations.**

The sequences of proteins are often encoded by initialising each amino acid position as a vector of 20 units in length, with each unit assuming a value of either 1 or 0 and each digit corresponding to a specific amino acid (a technique commonly used and referred to as one-hot encoding). To conduct diffusion on sequence space **Protein Generator** modifies this with each unit assuming a value between 1 and -1). In a given protein sequence, the true amino acid assumes a value of 1 while all others are denoted as -1. During the diffusion noising process Gaussian noise is subsequently applied to these units. Similar to the methodology of RFDiffusion, the authors employ RoseTTAFold as the denoising neural network, made possible by RoseTTAFold's capability to predict both sequence and structure from noised protein sequence data, a feature made possible by sequence masking during training. The loss function for Protein Generator is based on both the predicted sequence and structure. However, as diffusion takes place on sequence space, sequence classifiers can be integrated into the model during each denoising step, thus guiding the trajectory towards desired outcomes. This approach allowed the design of proteins with specified amino acid compositions, similarities to known families or folds, specific motifs scaffolded, and proteins capable of switching folds between two states when cleaved.

Structure representations.

An important recent development has been how to represent proteins to ensure output invariance to rigid transformations of the input data, such as rotation and translation. This is important so that a model can easily learn between proteins and so that a protein when rotated or translated is treated the same. There now exists many ways of achieving these different types of invariance or equivariance and so, first, we share some definitions provided by (Joshi, Mathis, Schmidt, Duval et al., 2023) to briefly define some terminology of the different groups of transformations: "1. $E(d)$: the Euclidean group includes translations, rotations, and reflections. It represents all possible transformations in d -dimensional Euclidean space. 2. $O(d)$: the orthogonal group represents rotations and reflections. $O(d) \leq E(d)$. 3. $SO(d)$: the special orthogonal group consists of rotations without reflections, in the d -dimensional space. $SO(d) \leq O(d)$. 4. $SE(d)$: the special euclidean group combines translations and rotations. $SE(d) \leq E(d)$. 5. $T(d)$: the translation group represents the symmetry transformations of pure spatial translations in d -dimensions, without any rotation or reflection. $T(d) \leq SE(d)$."

In order to achieve a representation of the protein that fits one of these groups, we can choose the set of parameters by which we define the nodes and edges of a protein graph and the operations we perform on those parameters. For example, Ingraham et al. were the first to introduce a series of vectors representing distance, direction, and orientation between two nodes in a protein graph that describe an $SE(3)$ representation of a protein upon which many papers now build. A particular helpful innovation they introduce is to describe the orientation between two nodes in a graph with a quaternion vector, which is more concise representation of rotation matrix. **RFDiffusion** employs this frame representation for each residue comprising its Ca coordinate and N-Ca-C rigid orientation. To noise the inputs during training, 3D Gaussian noise is applied to the Ca coordinates and Brownian motion is applied to the residue's orientation. **Genie** performs diffusion on Frenet-Serret frames (representations of the position and orientation of residues relative to a global reference frame) whilst **FrameDiff** performs diffusions on the rotations and translations (3D coordinates) separately, as in RFDiffusion, but uses an independent score network for each. This allows FrameDiff to operate using a loss of the difference between the true and predicted gradients of the denoising process. FrameDiff achieves similar performance to RFDiffusion with a quarter of the parameters — an important consideration for computational protein design where generative models are often run tens of thousands of times and the cost of running the model each time scales with its size.

Other representations (Contact maps and angles)

Both **Protein SGM** and **FoldingDiff** were earlier alternative diffusion approaches with different representations. ProteinSGM represents proteins as a series of matrices and recovered their structure using Rosetta whilst FoldingDiff models the backbone as a series of angles rather than 3D coordinates. This angle-based approach; however, is subject to 'lever' effects which refer to the phenomenon where small mistakes introduced at early stages of a process (in this case generating the backbone protein angles) can lead to large-scale alterations in the final outcome (protein structure). Consequently, RFDiffusion and FrameDiff have outperformed both ProteinSGM and FoldingDiff in designability.

and dynamically. For example, membrane transporters alternate between open and closed states, with gating facilitated by voltage-sensing domains. Progress has been made in structure-to-conformation models through machine learning predictions of molecular dynamics simulations, employing both convolutional neural networks [62] and diffusion models [63]. Initial sequence-to-conformation models include manipulating the MSA [64,65] and constructing diffusion models based on protein representations as a series of springs instead of coordinates [24]. However, these current conformational modelling approaches have achieved

limited success in efficiently sampling a protein's energy landscape. A key benchmark in future work should involve testing whether these models incorrectly predict multiple conformers for proteins that do not possess them.

Some initial success has also been achieved in designing conformations-to-structure workflows. Praetorius et al. [66] designed a new version of the eight alpha helical bundle designed-helical-repeat protein that can switch between two states, with one stabilised by a binding peptide. Protein Generator [37] was also used to design

proteins that change between different conformational states upon cleavage. The authors scaffolded the sequence of melittin, a pore forming protein, and a protease cleavage site to an additional 125 residues generated by the model, creating a protein whose pore forming capabilities are controlled until a protease cleaves off the new generated domain.

Discussion and future perspectives

Generative models' ability to sample from plausible protein space has enabled dramatic advanced across an exciting set of *de novo* protein design problems. However, the breadth of successes also highlights the remaining obstacles for protein designers. We propose understanding protein design through a framework highlighting five aspects of proteins important for protein design (Figure 3). What sequence of amino acids are they made of? What 3D structure do they fold into? How can this structure dynamically change shape? What does this protein structure do and what physiochemical properties does it have? And, finally, how is the protein's function regulated? The field of computational protein design has developed tools to connect these pieces of information: for example, protein structure prediction algorithms convert a 1D sequence into a 3D structure (sequence-to-structure), whereas inverse folding algorithms perform the opposite step (structure-to-sequence).

Key gaps highlighted by this framework include the inability to generate conformational landscapes of proteins based solely on sequence or structure, limited capacity to design proteins that undergo conformational changes, and incorporating structural regulation. Addressing these intricacies could pave the way for creating proteins responsive to small molecule ligands or for generating novel enzyme mechanisms. Devising these unique enzyme mechanisms might involve sampling multiple conformations that stabilise a transition state while maintaining a low affinity for the product.

Generative deep learning is currently limited by its ability to understand the physical interactions underpinning molecular behaviour. Most current approaches, which are trained on canonical amino acids, could encounter difficulties when expanding *de novo* design to include non-canonical amino acids. Current structure prediction algorithms often fall short in capturing the underlying physics of protein folding [67]. Water molecules are also often overlooked in protein design, despite being important for protein interactions, in particular for TCR interactions [68] and membrane pores [69]. These suggest that the performance of deep learning models in *de novo* design could be improved by introducing effective physics-inspired inductive biases.

Incorporating biochemical knowledge has been a key theme in improving generative models for protein design and engineering. Several studies have shown that incorporating biochemical insights in the training data, data processing and protein representation leads to significantly improved outcomes across different tasks. This was exemplified by Johnson [17], when removal of signal peptides in their data pre-processing resulted in non-functional proteins, and by Outeiral [29] who developed a language model trained on codons that outperformed larger protein LLMs. Moreover, biochemical knowledge in the form of wet lab data would further support the field. Robotic protein fitness landscape exploration tools combined with sequence based generative models open the opportunity for iterative rounds of active learning without scientist input [70].

Novel advances in deep learning are also likely to permeate *de novo* protein design. As the recent development of diffusion models rapidly resulted in the tools outlined above, it is likely that as more advanced models are developed in other areas, for example the recent development of consistency models [71], PGFM++ [72], or Geometric Neural Diffusion Processes [73] as alternatives to diffusion models, they will be translated into improvements for protein design. Indeed, the use of flow matching has already shown computational benefits over diffusion models without sacrificing accuracy [74,75]. An alternative approach might be the combination of models - combinations of LLMs and diffusion models have already been developed [16,38].

The developments outlined in this review also highlight that protein design is becoming increasingly accessible to a broader spectrum of researchers. This phenomenon is best exemplified by the release of the Chroma and RFDiffusion code and its accompanying interactive control suite. This democratisation promises to usher in revolutionary advancements, offering transformative solutions to pressing global challenges ranging from healthcare breakthroughs to environmental remediation. Sectors with low market barriers, such as manufacturing, plastic degradation, and carbon capture, are likely to see initial breakthroughs. However, as the field progresses, it is imperative that researchers remain aware of potential biosecurity risks. A judicious and proactive approach will be vital to ensure that the advancements in *de novo* protein design are harnessed safely.

We envisage protein design evolving not only into a staple research tool for developing binders and control proteins but also as a key instrument to address and resolve grand challenges.

Declaration of competing interest

Adam Winnifrieth is a founder and holds equity of Evolvere Biosciences Ltd. B.L.H. acknowledges outside interest in Prox Biosciences as a scientific cofounder. Carlos Outeiral declares no conflict of interest.

Data availability

No data was used for the research described in the article.

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